Lecture 1: Introduction to Port Engineering

Port Engineering Course

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Why Ports Matter – Engineering & Economics

- 80–90% of global trade transported by sea
- Ports enable bulk commodities, energy, vehicles, and logistics systems
- Critical nodes in national economic resilience
- Infrastructure decisions directly affect competitiveness

Marine Environment & Design Parameters

- Significant wave height (Hs) & return period analysis
- Tidal levels: MSL, HHWL, LLWL
- Storm surge & extreme water levels
- Sediment transport & estuarine dynamics

Container vs Non-Container Terminals

- Container terminals handle standardized TEU units using cranes and yard stacking systems
- Non-container terminals handle dry bulk, liquid bulk, general cargo, and vehicles
- Dry bulk: coal, grain, clinker – requires conveyor & stockyard systems
- Liquid bulk: fuel, CPO – requires pipelines, storage tanks, safety buffer zones
- Car terminals (Ro-Ro): ramps, circulation lanes, large parking yards
- Infrastructure design differs significantly between terminal types

Diagram: Marine Forces → Infrastructure Response

Terminal Engineering Fundamentals

- Dry bulk: berth length, conveyor systems, stockyard layout
- Liquid bulk: jetty structures, pipelines, safety zones
- Car terminals: Ro-Ro ramps, yard circulation efficiency
- Interface between infrastructure & equipment productivity

Diagram: Simplified Bulk Terminal Layout

Applied Perspective: Bulk Terminals in Practice

- Dry & liquid bulk dominate national revenue streams
- Terminal specialization influences market share
- Sedimentation & draft limitations affect competitiveness
- Infrastructure investment must align with strategic priorities

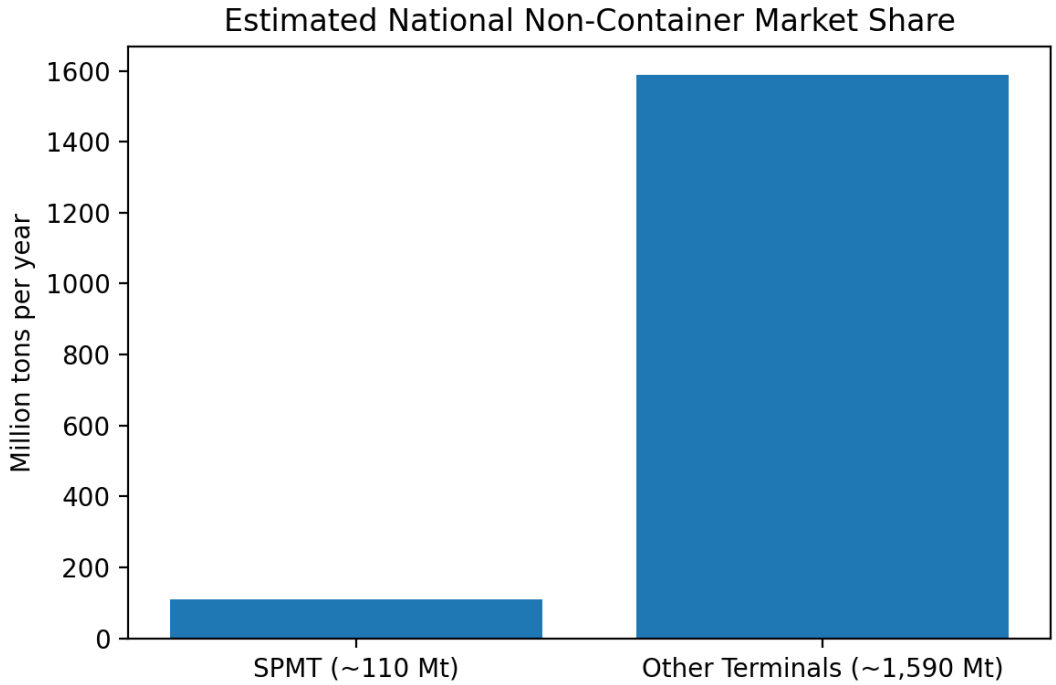
Course Roadmap

- Part I: Marine & Physical Infrastructure
- Part II: Terminal Engineering Systems
- Part III: Performance & Strategic Considerations
- Integration of engineering, operations, and governance

Closing Reflection

- Port engineering is not only about structures.
- It is about enabling national connectivity, commodity flow, and long-term resilience.
- Different terminal types require different engineering solutions.

SPMT Estimated Market Share (Non-Container)



Assumptions & Calculatio

- Total national non-container ≈
- SPMT throughput ≈ 110 millio

Market Share ≈ 110 / 1700 = 6.5

Strategic Implications:

- Significant growth opportunity
- Non-container sector largely o
- Infrastructure & governance re

Slide 7: Comparative Terminal Handling Diagrams

Comparison of Terminal Handling Flows (Simplified Engineering Logic)

Bulk:	Berth	→ Conveyor →	Stockyard
Container:	Berth (STS)	→ Yard (RTG/RMG) →	Gate
Liquid Bulk:	Berth (Loading Arm)	→ Pipeline →	Storage Tanks
Car (Ro-Ro):	Ramp Berth	→ Parking Yard →	Gate / Distribution